

i n v e n t o r y s

I/A Series Hardware Overview

I/A Version 8.3
HPS

I/A Series Equipment Overview

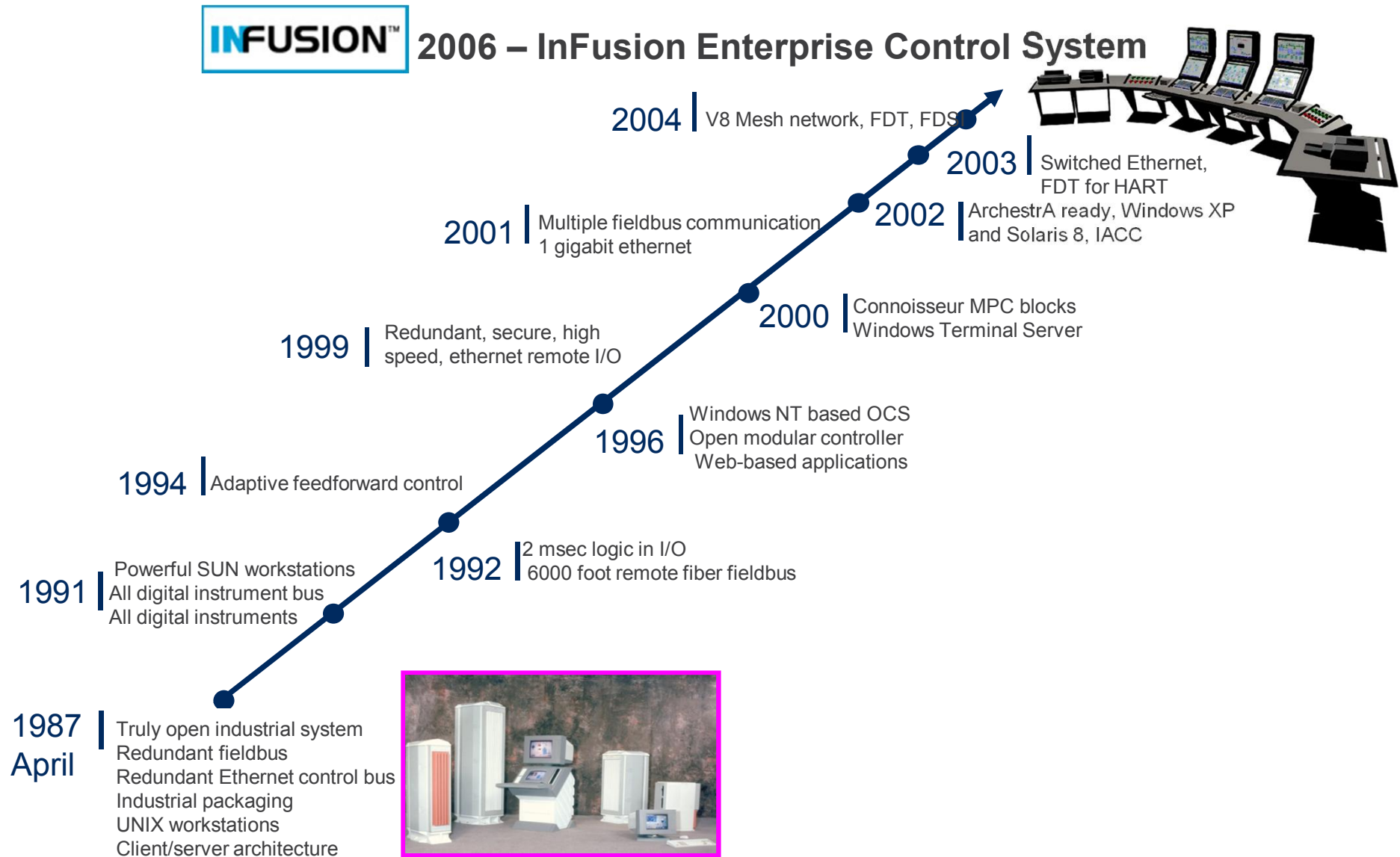
Module Objective:

- List the basic hardware components of an I/A Series system functions and how they work with each other

Enabling Objectives:

- Identify and describe the major components and functions
 - Control Processors and Fieldbus Modules
 - Workstations
 - Application Processors
- Describe The Mesh control network architecture.
- Use the Letterbug Configurator

I/A Series Evolution



Major Components

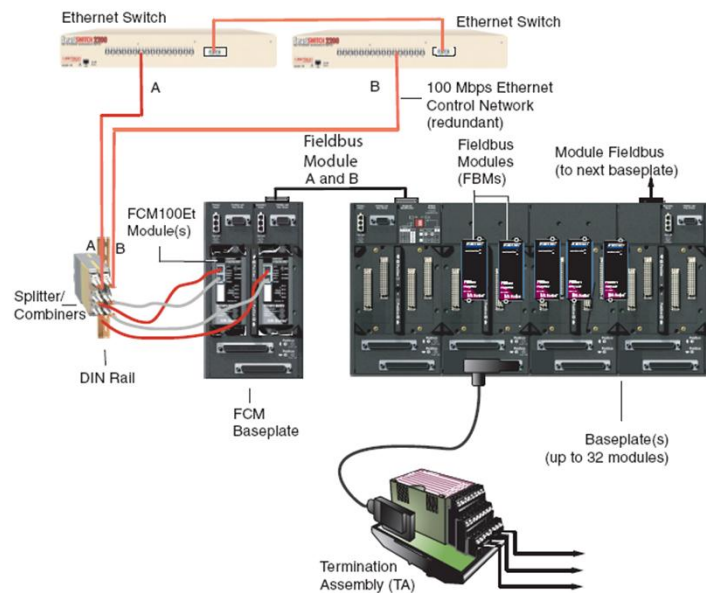


- Module types
 - Control Processors and Fieldbus Modules
 - Workstations
 - Application Processors
- Mesh network
- Distributed Open System

Module Types

- Processor Modules (Stations)
 - **Control Processor**
 - **FCP270**
 - **Servers & Workstations**
- Field Modules
 - **200 Series Fieldbus Modules (FBMs)**
- Ethernet Switches
 - **16, 24, 48 and 96 Port**

Fieldbus Modules (FBM) – What do they do?



- **Converts source signals to I/A digital signals**
 - Transmitters (analog & digital)
 - PLCs,
 - third party devices
- **Converts I/A digital signals to final device signals**
 - Valves
 - Motors, pumps
- **Determines resolution of signal**
- **Fail-safe values for final device**
- **Optional**
 - Runs Ladder logic
 - Sequence of events
- **Status Indicators for discrete signals**

Fieldbus Modules – FBM Types

- **Analog Signal Conversion**

- 0-20 mA input and output
- 0-10 Vdc input and output
- Thermocouple (T/C) input
- Resistance Temperature Detector (RTD) input
- Run Transient Data Recorder (TDR) / Analyzer w/ 10 ms resolution

- **Discrete Signal Conversion**

- Contact sense
- Voltage monitor
- Pulse (flow totalization) input
- Ladder logic
- Sequence of Events (SOE) w/ 1 ms resolution

Analog FBMs

Fuqing and Fangjiashan FBMs

FBM201	8 Analog In, 0-20 mA dc
FBM201b	8 Analog In, 0-100 mV dc
FBM201c	8 Analog In, 0-5 V dc
FBM201d	8 Analog In, 0-10 V dc
FBM202	8 Thermocouple/mV In, -10.5 to 69.5 mV dc
FBM203	8 RTD In, 0-320 ohms
FBM203b	8 RTD In, 0-640 ohms
FBM203c	8 RTD In, 0-30 ohms
FBM204	4 In (0-20 mA dc), 4 Out (0-20 mA dc)
FBM205	4 In (0-20 mA dc), 4 Out (0-20 mA dc), Redundant Ready
FBM206	8 Pulse In, 10 Hz to 25 kHz
FBM208	Redundant with Readback; 4 In & 4 Out (0-20 mA dc)
FBM211	16 Analog In, 0-20 mA dc
FBM212	14 Thermocouple/mV In, -10.5 to 69.5 mV dc
FBM237	8 Analog Out, 0-20 mA dc, Redundant Ready

Digital FBMs

Fuqing and Fangjiashan FBMs

FBM207	16 In, 0-80 V dc
FBM207b	16 In, 24 V dc Contact Sense
FBM207c	16 In, 48 V dc Contact Sense
FBM217	32 discrete input
FBM219	24 inputs, 8 outputs Voltage monitor/contact sense
FBM240	8 discrete inputs, 8 discrete outputs (Redundant)
FBM241	8 discrete inputs, 8 discrete outputs
FBM242	16 discrete outputs

Fieldbus Modules – FBM Types

- **Field Communications Protocols**
 - FoxCom input and output
 - HART input and output
 - FOUNDATION Fieldbus input and output
 - Profibus-DP input and output
 - Modbus input and output
- **Field Device System Integrator (FDSI) Communications**
 - RS-232, RS-422, RS-485 inputs and outputs
 - Ethernet inputs and outputs

FBM - Field Communications

Fuqing and Fangjiashan have FBM 230/231 and 233 (FDSI)

FBM214/215	HART
FBM216	HART Redundant Ready
FBM220/221	FOUNDATION Fieldbus H1
FBM223	Profibus
FBM224	Modbus Interface
FBM228	FOUNDATION Fieldbus H1 Redundant
FBM243/246	FoxCom (246 is Redundant)

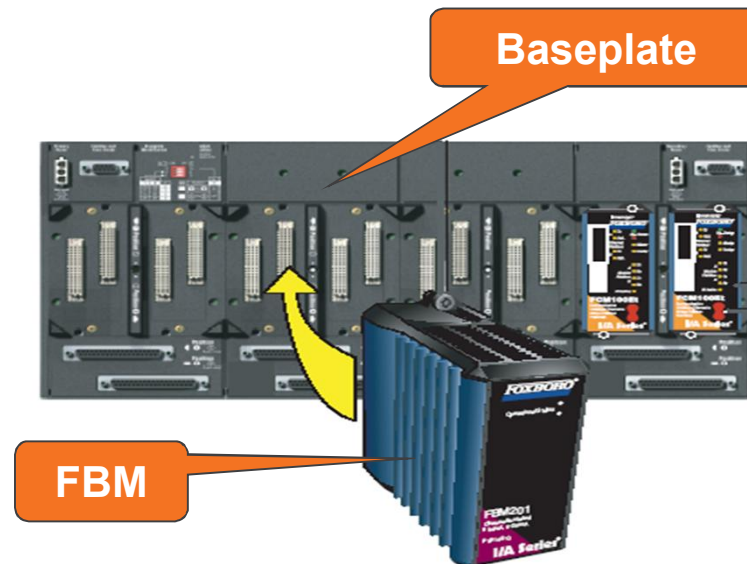
FDSI

FBM230/231 **RS232/422/485:Modbus, AB, etc. (231 is Redundant)**

FBM232/233 **Ethernet:Modbus, OPC, etc. (233 is Redundant)**

Fieldbus Modules - How are they installed?

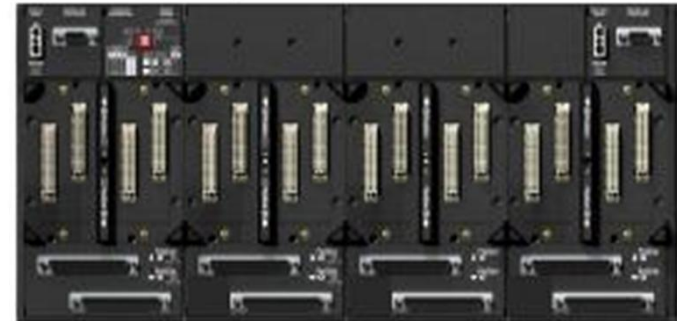
- **200 Series Baseplates**
 - Horizontal
 - Vertical



How are FBMs Installed?

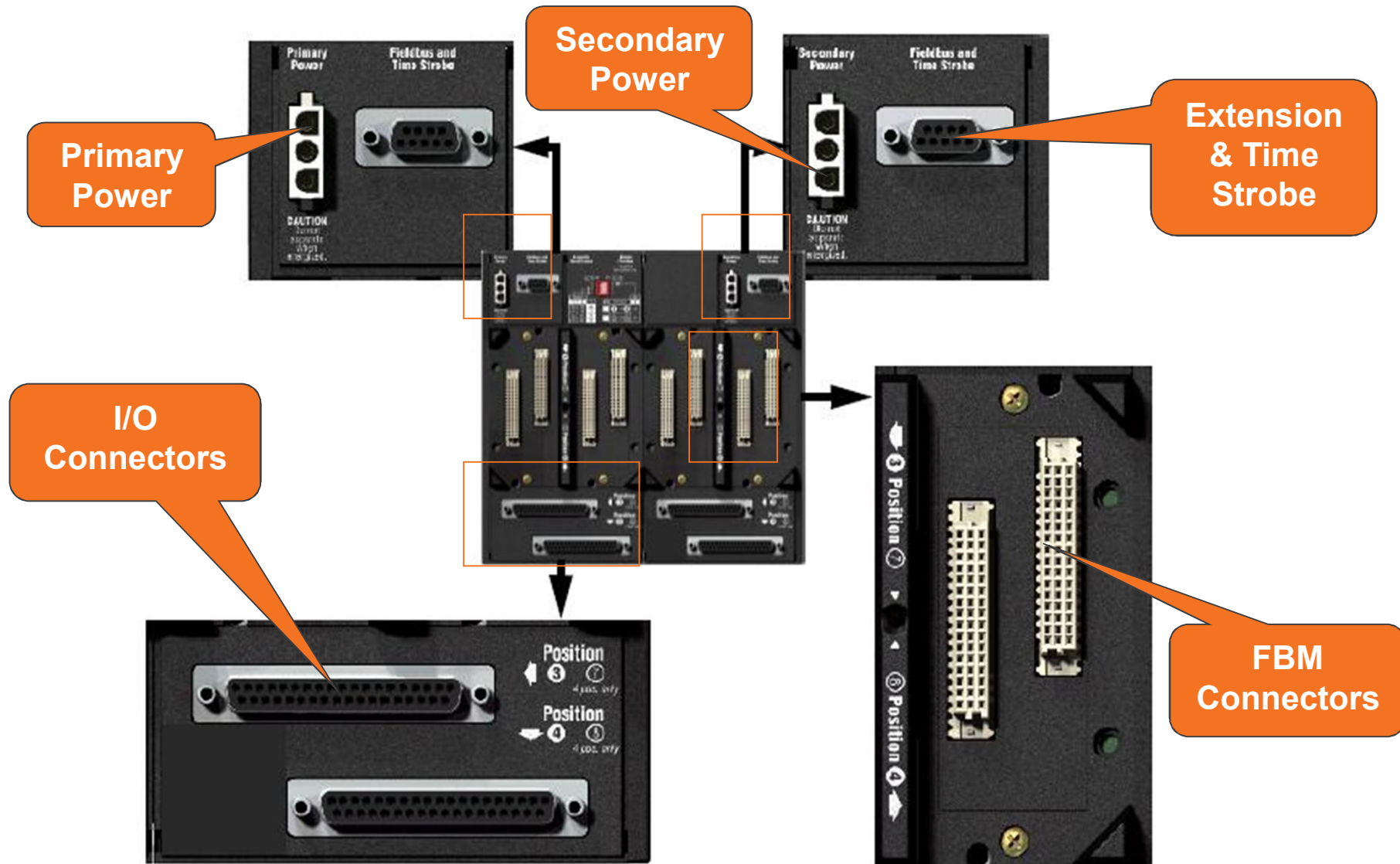
Modular Baseplates

- **DIN rail mounted**
 - FCP270
 - FBMs
- **Sizes available**
 - 2, 4, 8 unit module positions
 - Horizontal or Vertical
- **Power connectors for redundant power**
- **Redundant Fieldbus**
- **Time strobe and I/O signal cabling**



How are FBMs installed?

Baseplate connections

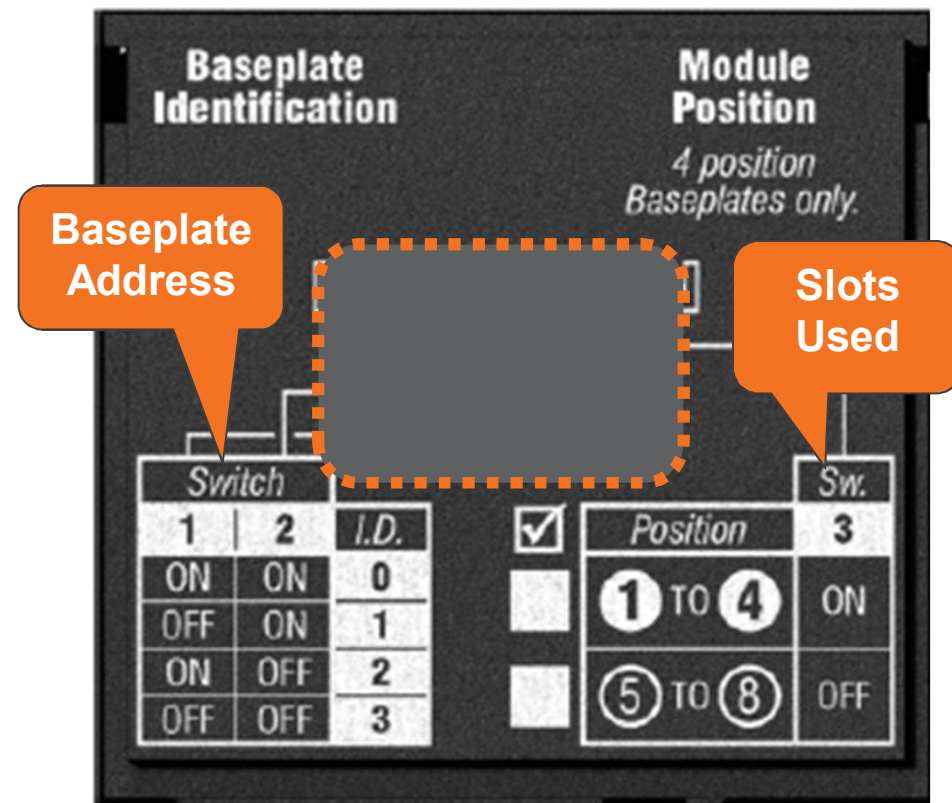


How are FBMs identified?

200 Series

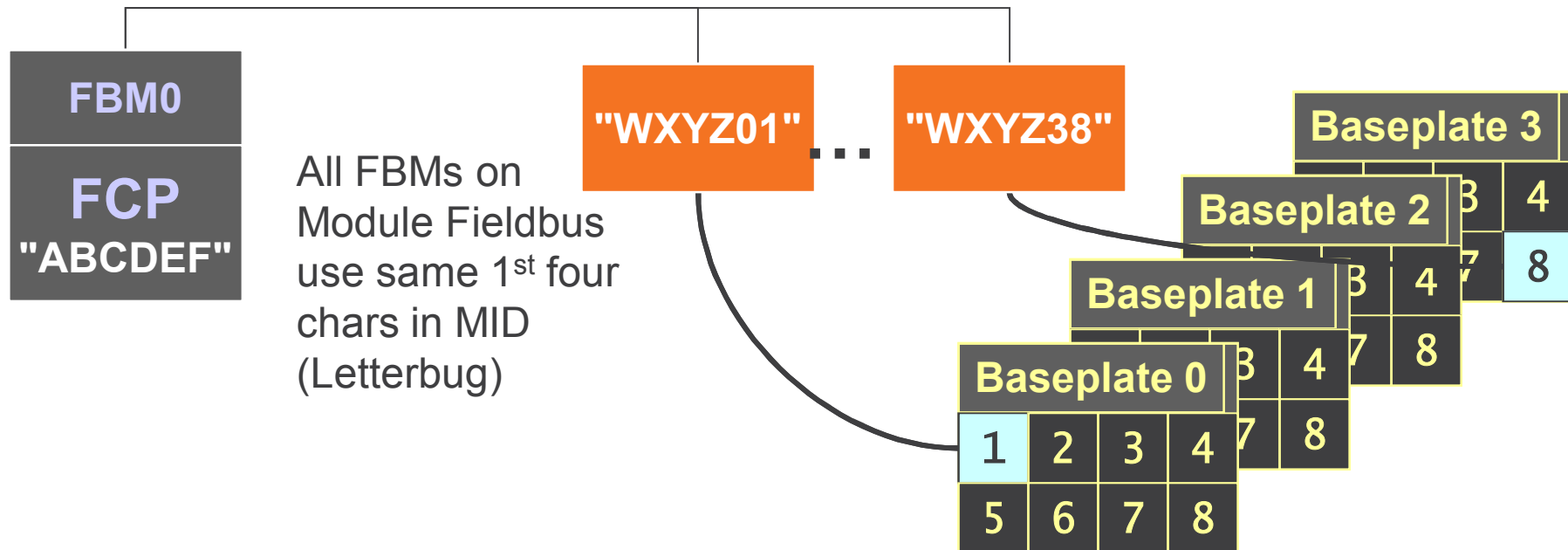
- **Module Identifier (MID) for FCP270 entered using infrared, hand-held Letterbug Configurator**
- **6-character MID for FBM**
 - Character 1-4
 - User-defined
 - Character 5
 - Baseplate # (0..3)
 - Set by switches 1&2
 - Character 6
 - Physical slot # in baseplate (1..8)

A B C D _ _



How are FBMs identified?

200 Series basic configuration



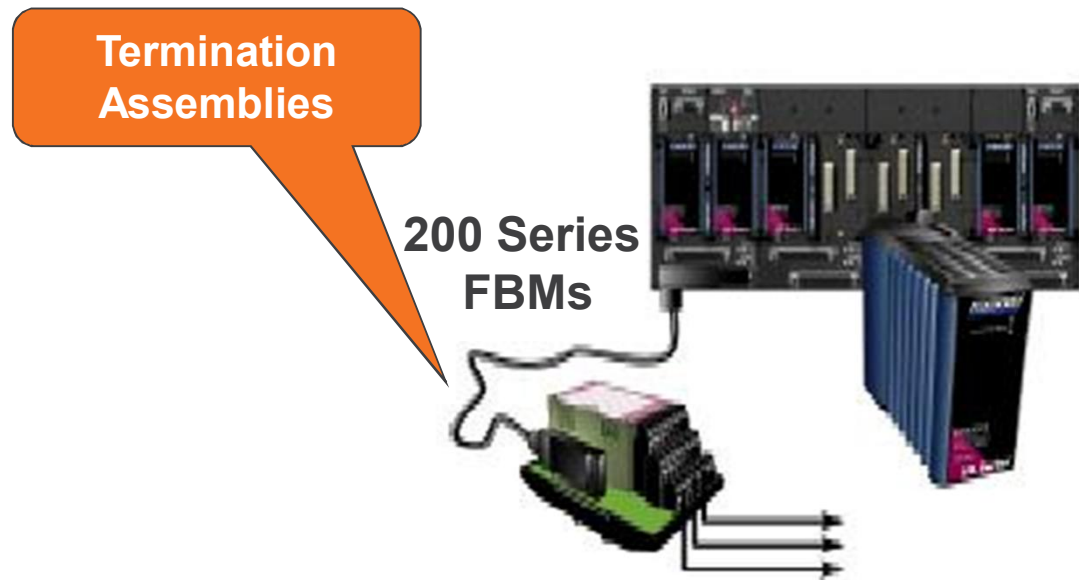
FBM Status Indicators

Red LED	Green LED	Status
Off	Off	Power to card failed or switched off.
On	Flashing	Failure detected during start-up diagnostics
On	On	Start-up diagnostics passed and FBM is ready to be brought on-line by the CP.
Off	On	FBM on-line and functional. (This is the normal “run” state.)
Off	Flashing	FBM in fail-safe mode (communication errors with CP)

B0400FA - DIN Rail Mounted FBM Subsystem User's Guide

Termination Assemblies - What do they do?

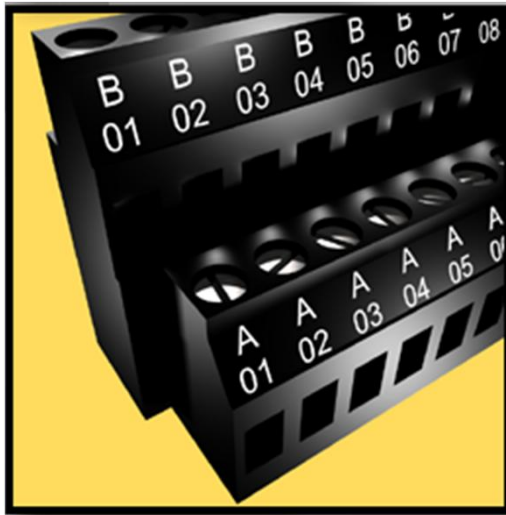
- **Field Signal Connection**
- **Links FBM with field devices**
 - Different TA's available for each FBM
 - Passive allows direct field signal feed to FBM
 - Active signal conditioning separated from FBM



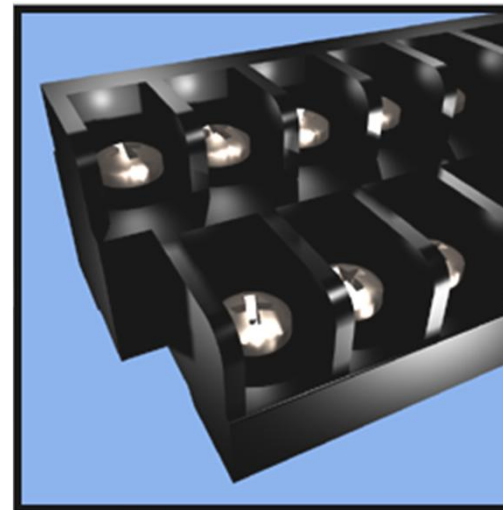
Termination Assemblies - Wiring Connections

- Termination Assembly types
 - Analog – **Magenta**
 - Discrete – **BLUE**
 - Field communication – **GREEN**
- Termination Assembly connection types

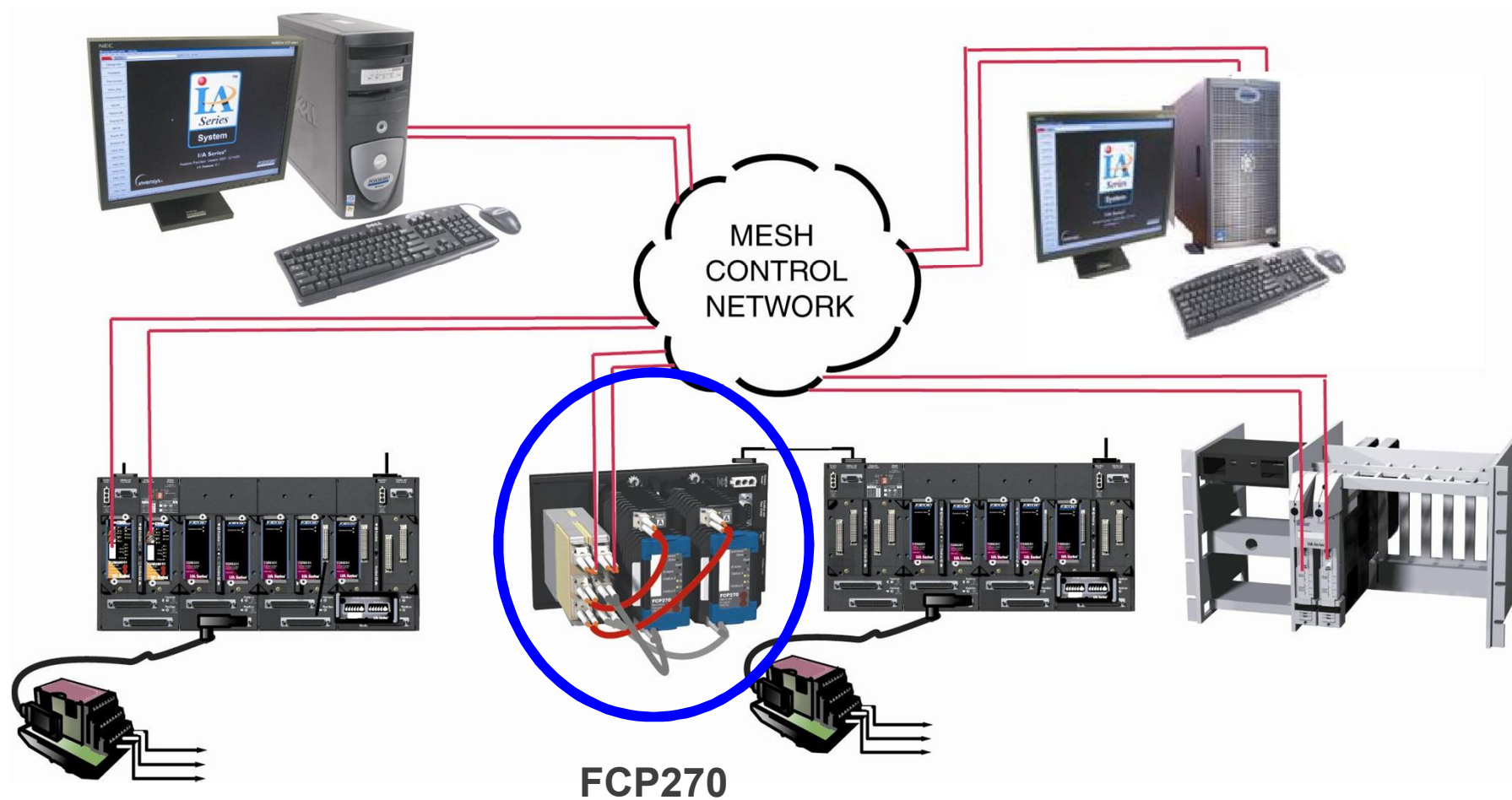
Compression Screw



Ring Lug



I/A V8 Major Components - Control Processors



Control Processors - FCP270



FCP270

Control Processor FCP270 Features

- DIN Form factor
- Hardened field-mounted control
- Talks directly to Fieldbus Modules
- **Performs**
 - Regulatory, logic, timing, and sequential control
 - Data acquisition and alarm detection and notification
 - 4000 control blocks maximum

FCP270



Control Processor - Features

- **Optional global positioning – time sync**
- **Infrared interface to I/A Series Letterbug Configurator**
- **Memory resident image - reboot time for database <10 seconds**
- **Single or Fault Tolerant**
 - FT modules installed adjacent

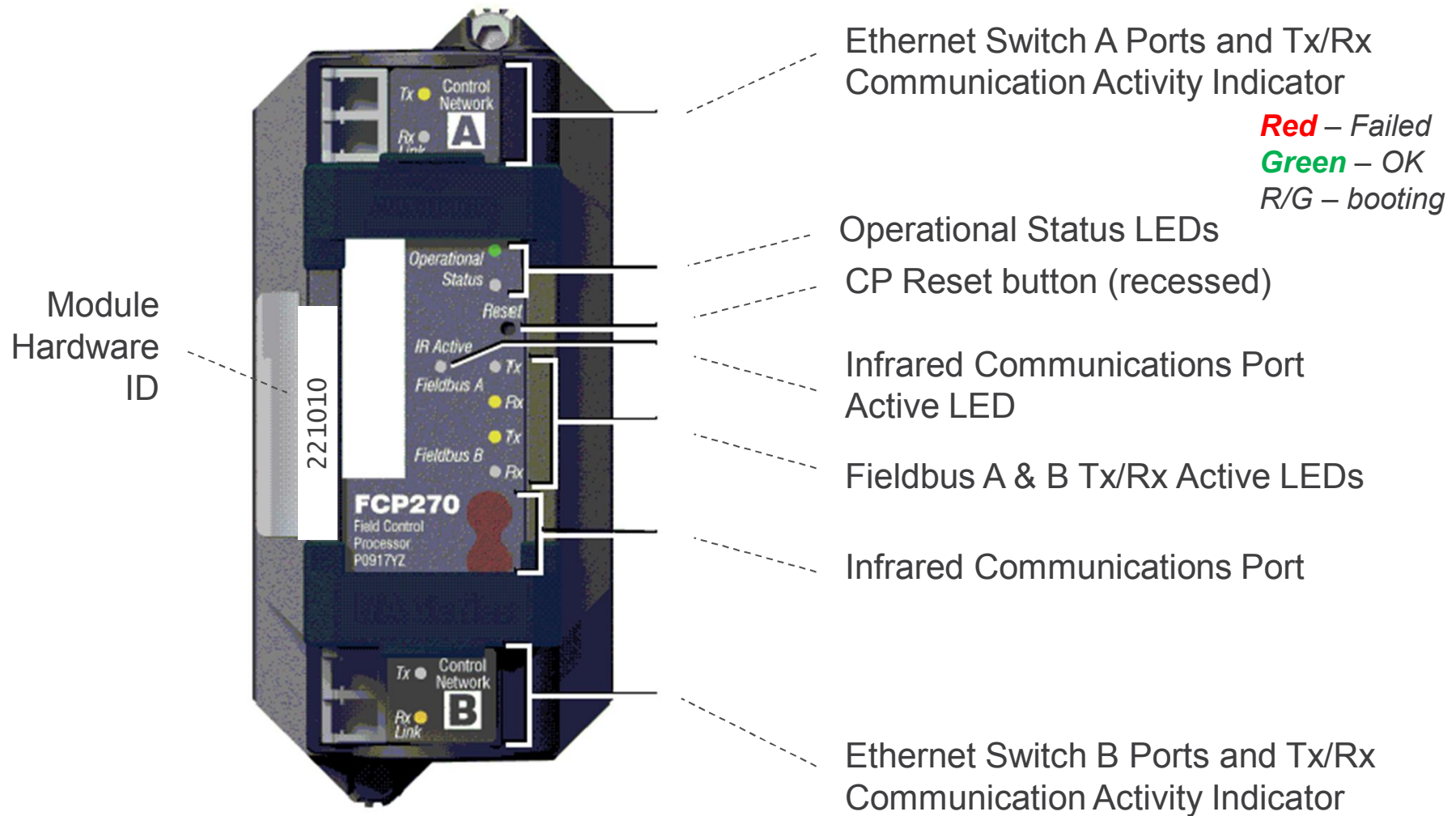
Control Processors

What hardware is supported?

FCP270

- 32 x 200 Series FBMs maximum
- Field Device System Integrator (FDSI)

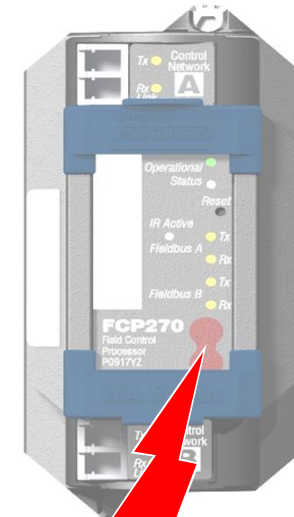
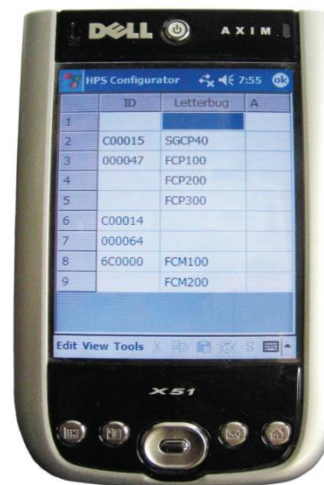
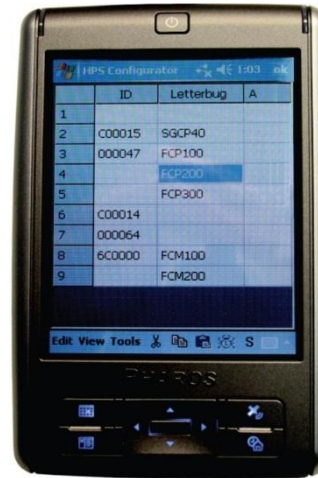
FCP270 Front Panel



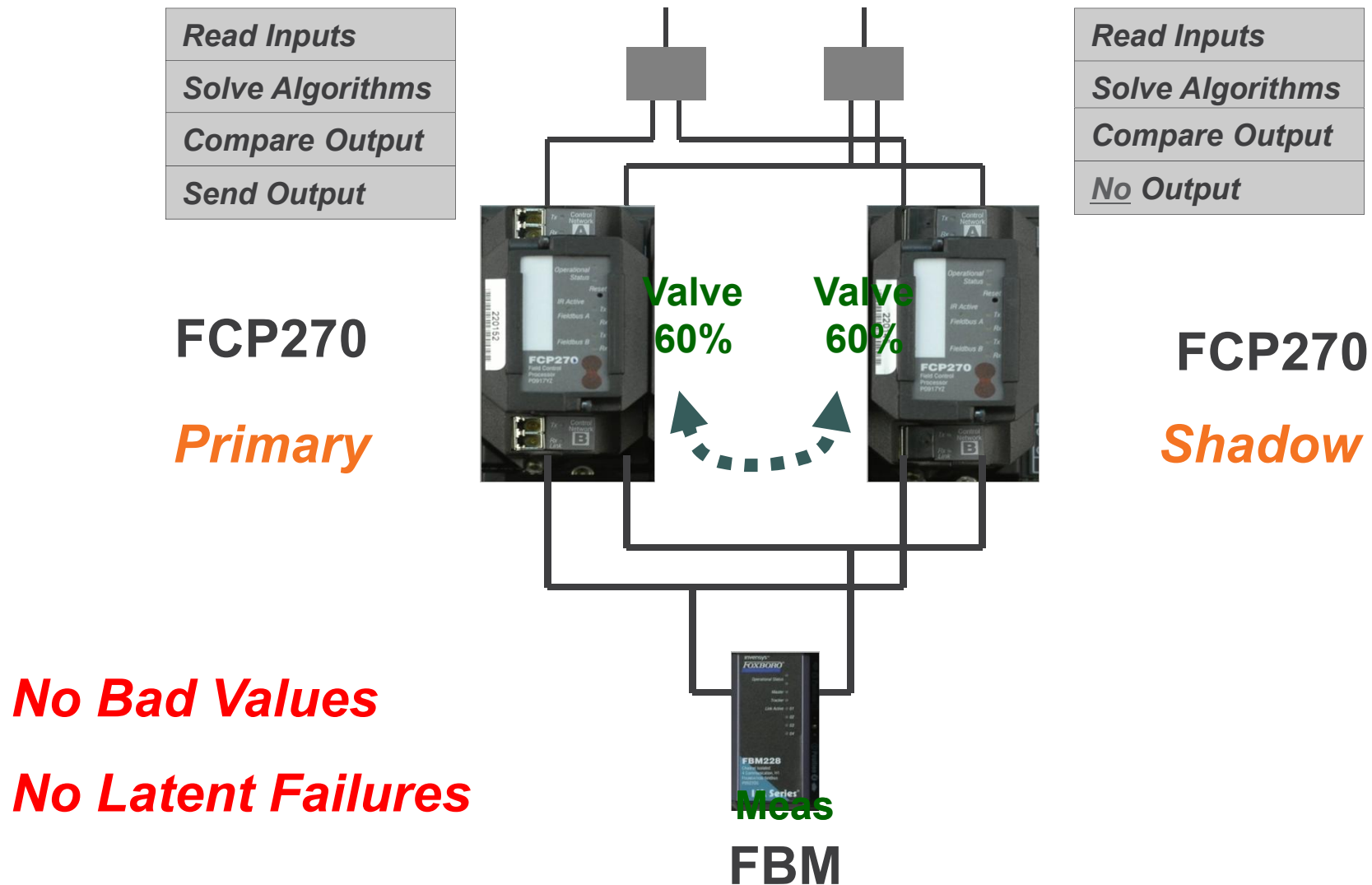
Configuring Module Identifier (MID)

Letterbug configurator - Model B21

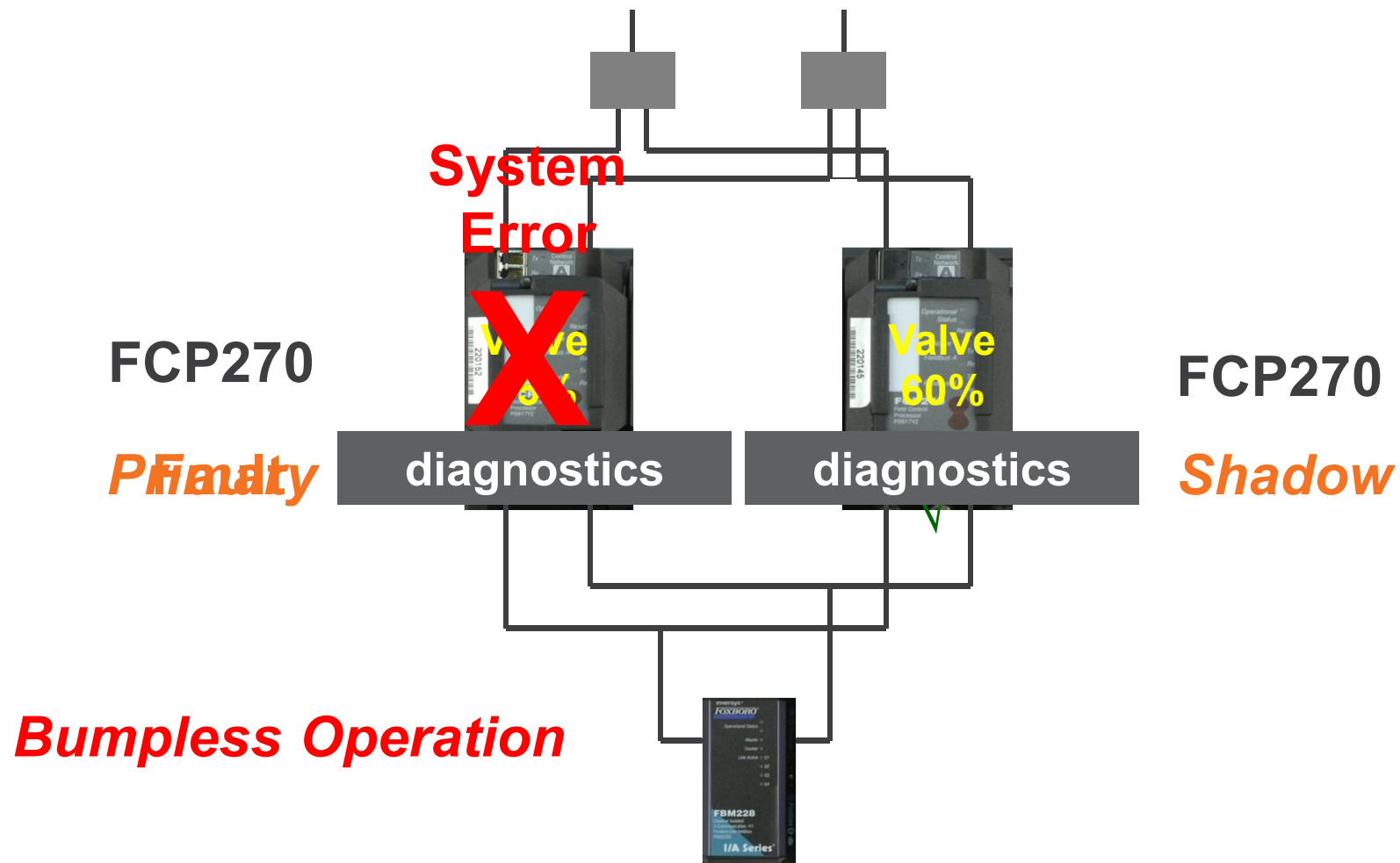
- Configured using IR communications
 - FCP270
- Supported units
 - Pharos 535e/525e
 - Dell Axim X51
 - HP iPAQ 2200



Unique & Patented - I/A Fault Tolerance



Unique & Patented - I/A Fault Tolerance



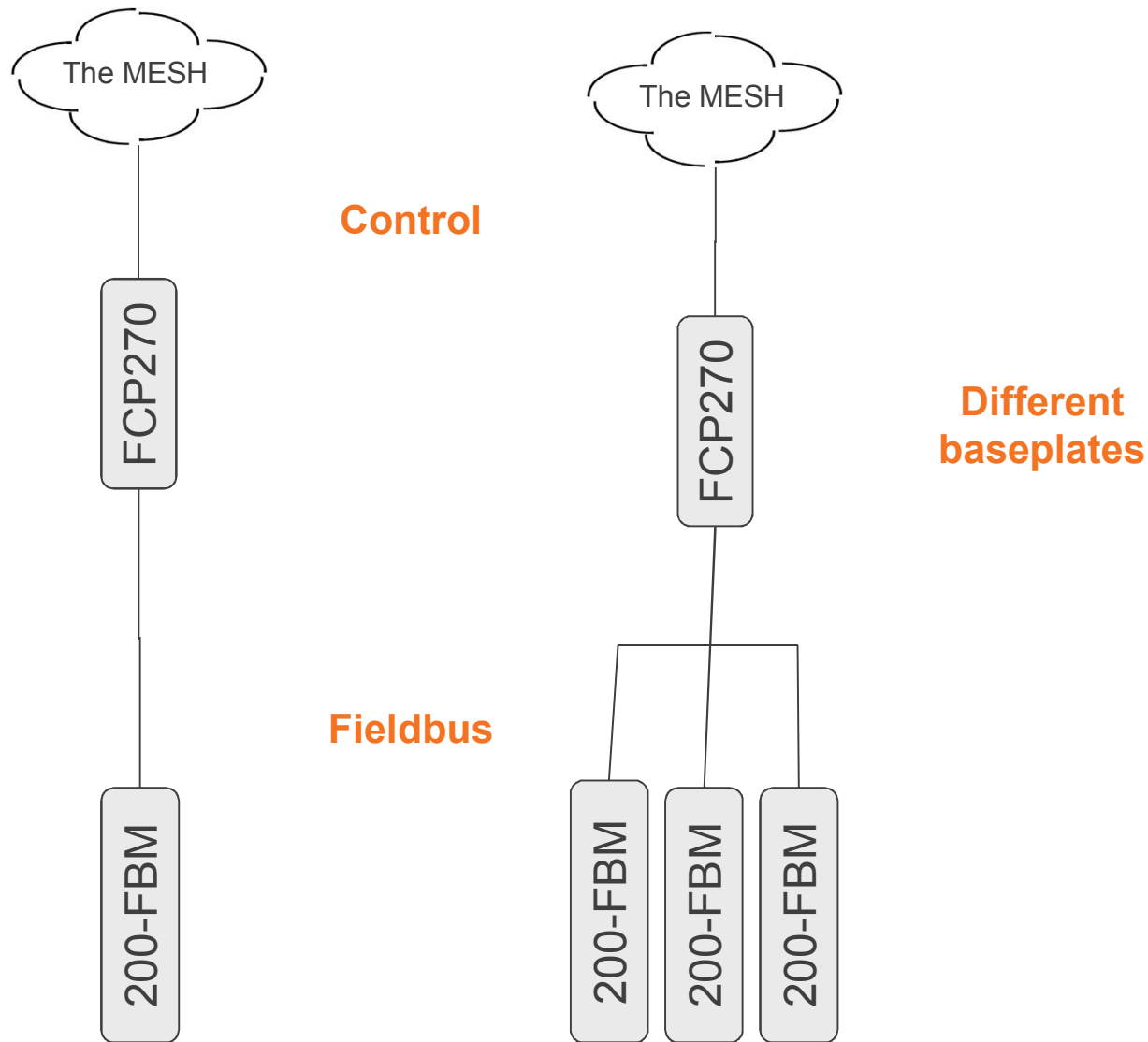
Control Processor - FCP270

- Similar in size to the FBM
- Also mounted on a baseplate
- Provides process control and serves as a communication interface between the FBMs and the MESH Control Network
- FCP270 can be a single or Fault Tolerant module
Fuqing and Fangjiashan are all Fault Tolerant

The FCP270 can communicate with ...

- **200 Series FBMS via a 2 Mbps Module Fieldbus**
 - Outputs a 268 Kbps output signal identical to the typical 268 Kbps source from the FCP270

Control Processors - FCP270 Talking to FBMs



Control Processors

FCP270 Installation Sizing Guidelines - Cable Lengths

- **Station to MESH, 100 Mbps**

- Ethernet switch to Splitter Combiner/FCP270 total 2 km (1.24 mi) maximum

- **Module Fieldbus - 2 Mbps**

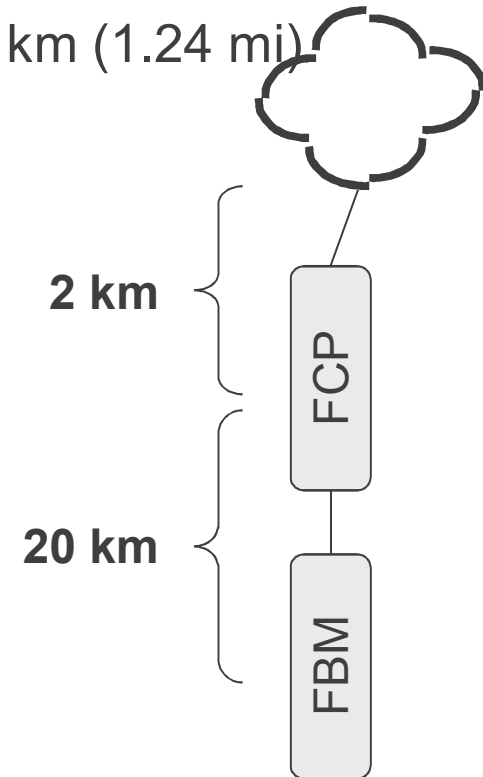
- Shielded twisted-pair segment
 - Between two baseplates: 60 m (198 ft) maximum
 - Overall 60 m (198 ft) maximum

- **Fiber optic segment**

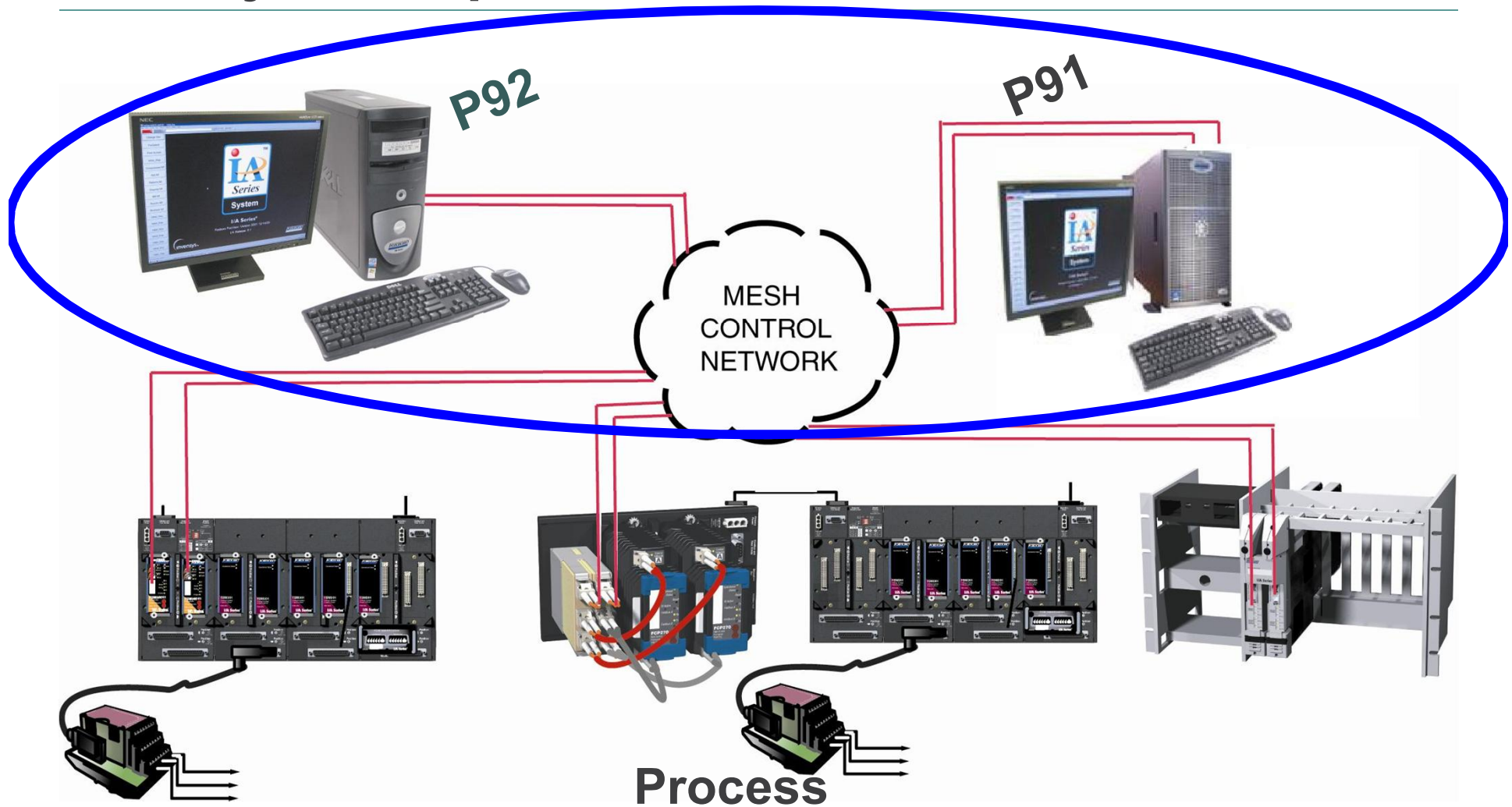
- Between two baseplates: 10 km (6.2 mi) maximum
- Overall 20 km (12.4 mi) maximum

- **Combined End-to-End (using coaxial and fiber optic cabling)**

- Control station to farthest FBM, 20 km (12.4 mi)



I/A Major Components - Servers & Workstations

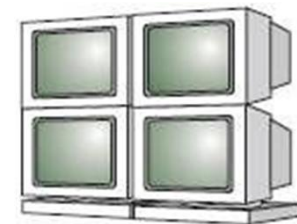


Workstation Server - Model P91 Overview

- **Server Class Machine**

- **Supports**

- I/A Configuration Tools
 - IACC
 - ICC (Default)
 - AIM*
- Operations
- Remote client access to I/A applications
- Multi-screen capability
- FoxView



Workstations Server - P91 Base Configuration

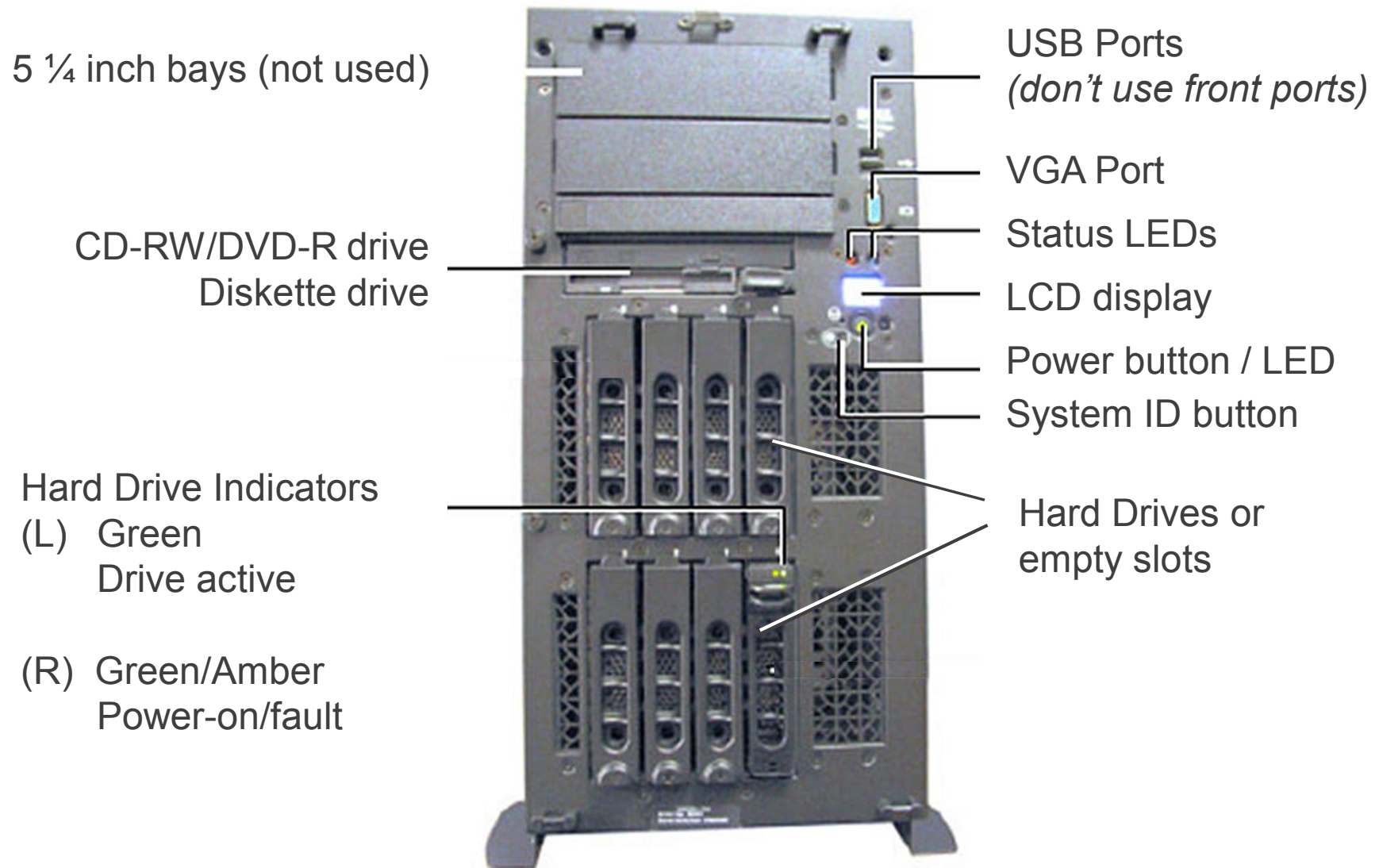
- Intel Xeon Processor, 2.83 GHz
- 1 GB ECC DDR-2 SDRAM (expandable to 4 GB)
- 1 Serial interface (printer)
- Internal 73 GB hard drive
 - non-RAID, **Serial Attached SCSI (SAS)** interface
- Internal EIDE CD-RW/DVD drive
- Internal 3.5-inch, 1.44 MB disk drive
- 2 Integrated 10/100/1000BaseT Ethernet ports
- USB mouse, keyboard
- Redundant hot-swap power supplies
- Four rear USB ports



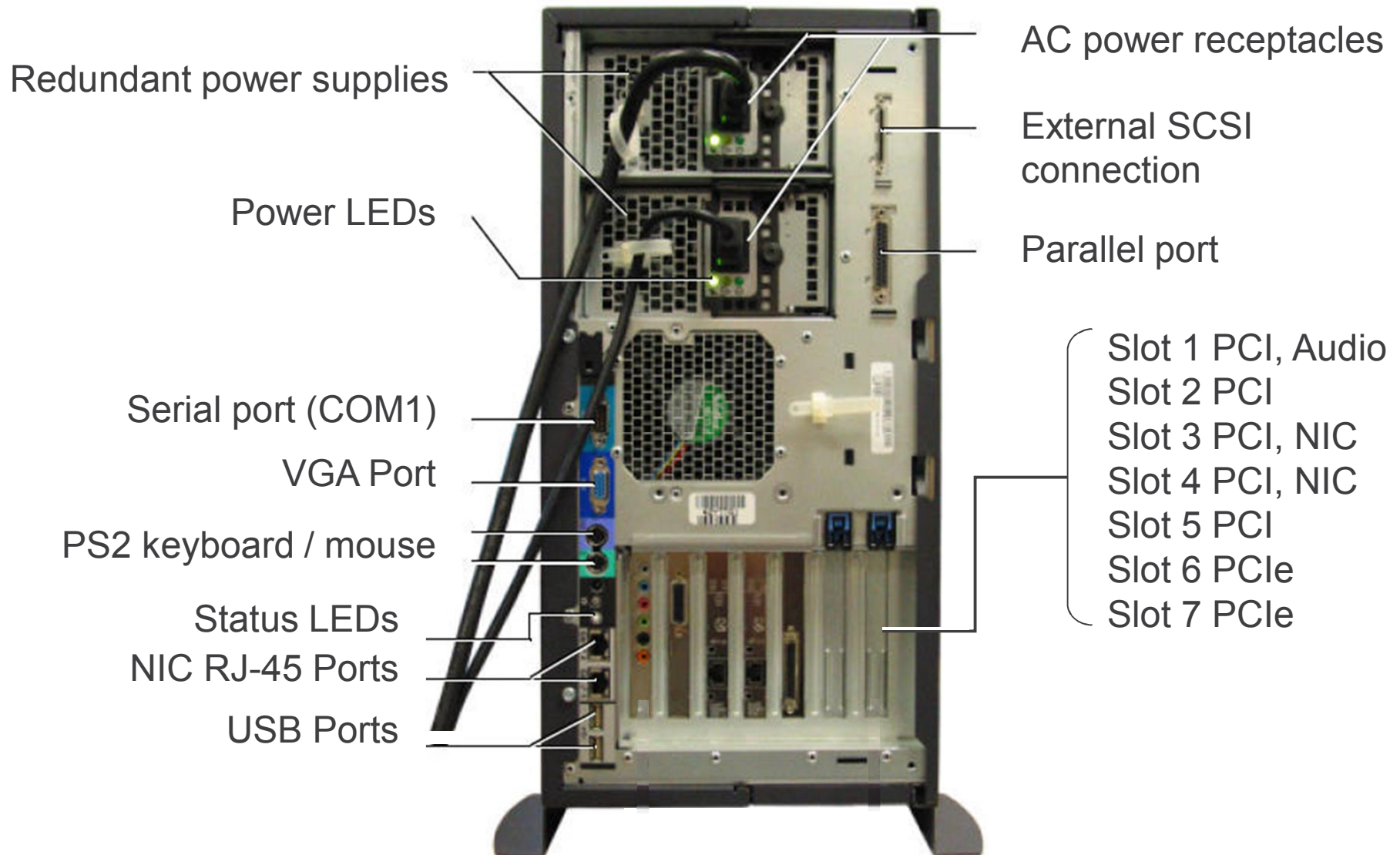
Workstations Server - P91 Options

- Expandable to eight hard drives, either 73 GB or 300 GB
- Monitors with optional USB touchscreens
- Multi-port serial/parallel card
- Dual or quad monitor graphics card
- RAID1 or RAID5 internal hard drive arrays
- External AIT 80/208 GByte tape drive
- Ethernet adapter cards (copper or fiber)

Workstations - P91 Front Panel Layout



Workstations - P91 Rear



Workstations Server – Model P90

- Intel Xeon Processor
- 1 GB ECC DDR-2 SDRAM (expandable to 4 GB)
- 1 Serial interface (printer)
- Internal 73 GB hard drive
 - non-RAID, Serial Attached SCSI (SAS) interface
 - Optional RAID1 or RAID5
- Internal CD-RW/DVD drive, 3.5-inch, 1.44 MB disk drive
- 2 Integrated 10/100/1000BaseT Ethernet ports
- USB mouse, keyboard
- Redundant hot-swap power supplies
- 2 rear, 2 front USB ports



Workstations - Model P92 Overview

- Workstation
- Supports
 - Operations
 - FoxView
 - Engineering Environments
 - Multi-screen capability

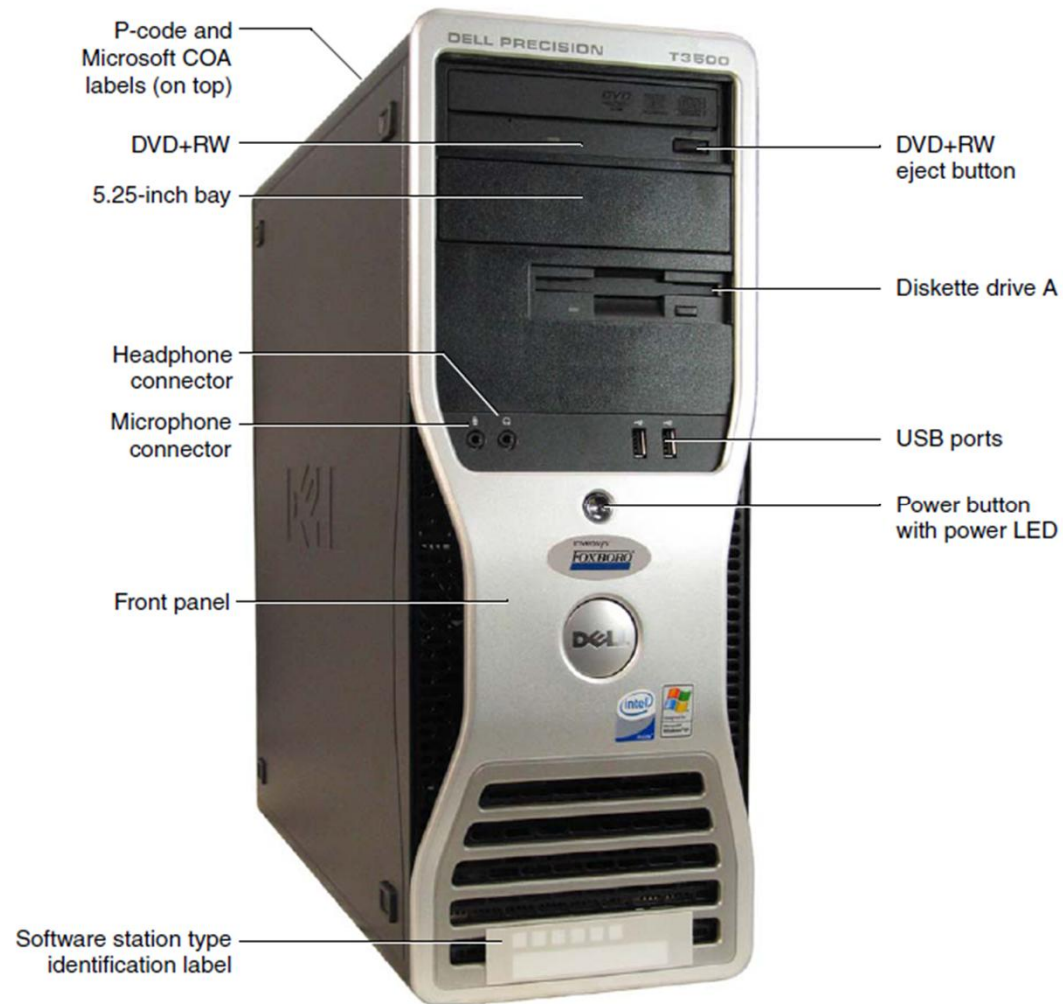


Workstations - Model P92K Configuration

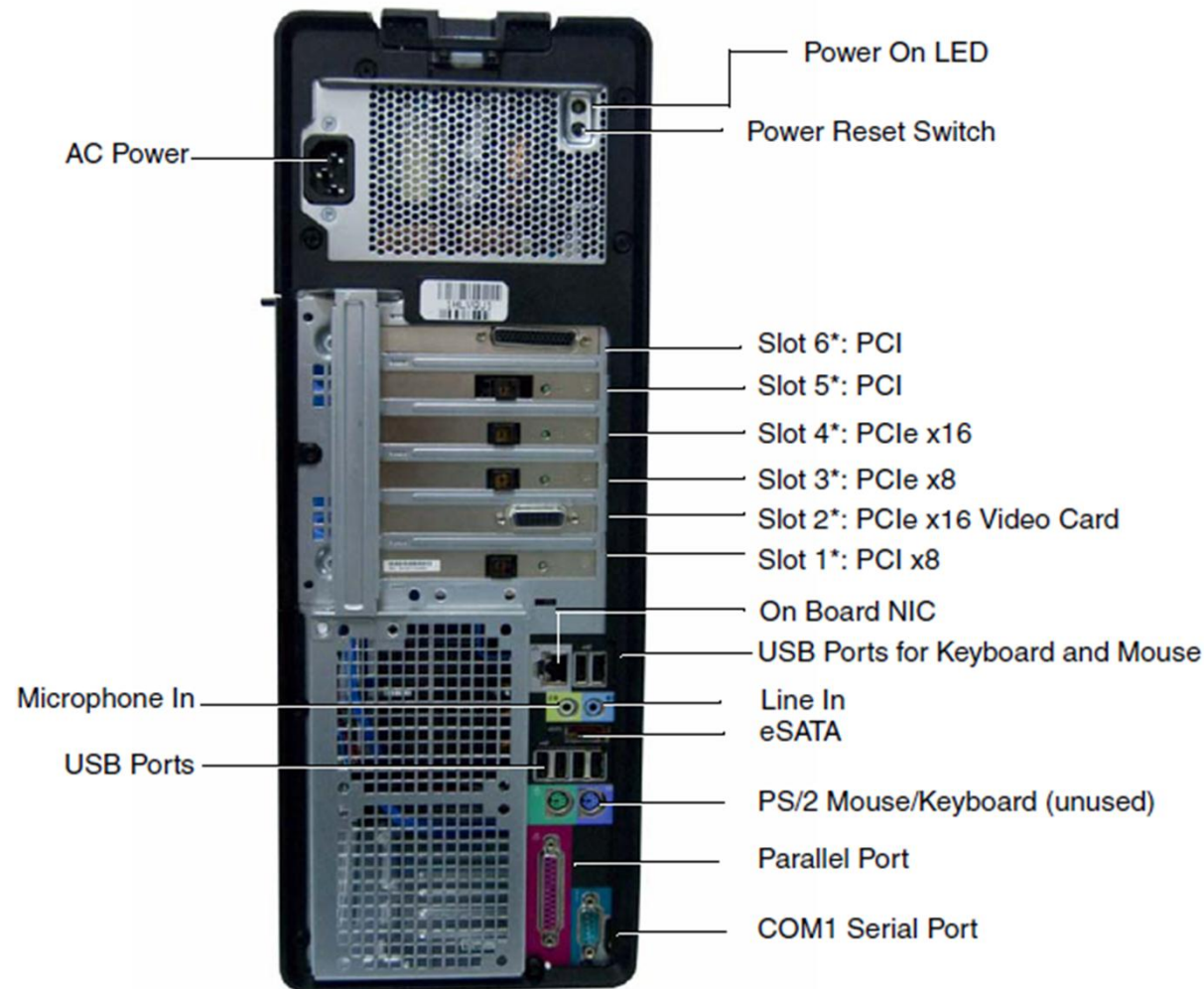
- Intel® Xeon™ W3520 quad-core 64-bit processor (2.67 GHz)
- 3 GB DDR3, 1066 MHz
- 250 GB SATA hard drive (expandable to 500 GB), RAID 1 optional
- Parallel interface port for printer
- Internal, 3.5-inch 1.44 MB floppy disk drive
- PCI Express™ x16 video card
- A serial interface port for:
 - Printer
- Internal DVD+RW/CD drive
- Dual monitor graphics card
- Integrated 10/100/1000BaseT Ethernet port
- USB mouse and keyboard



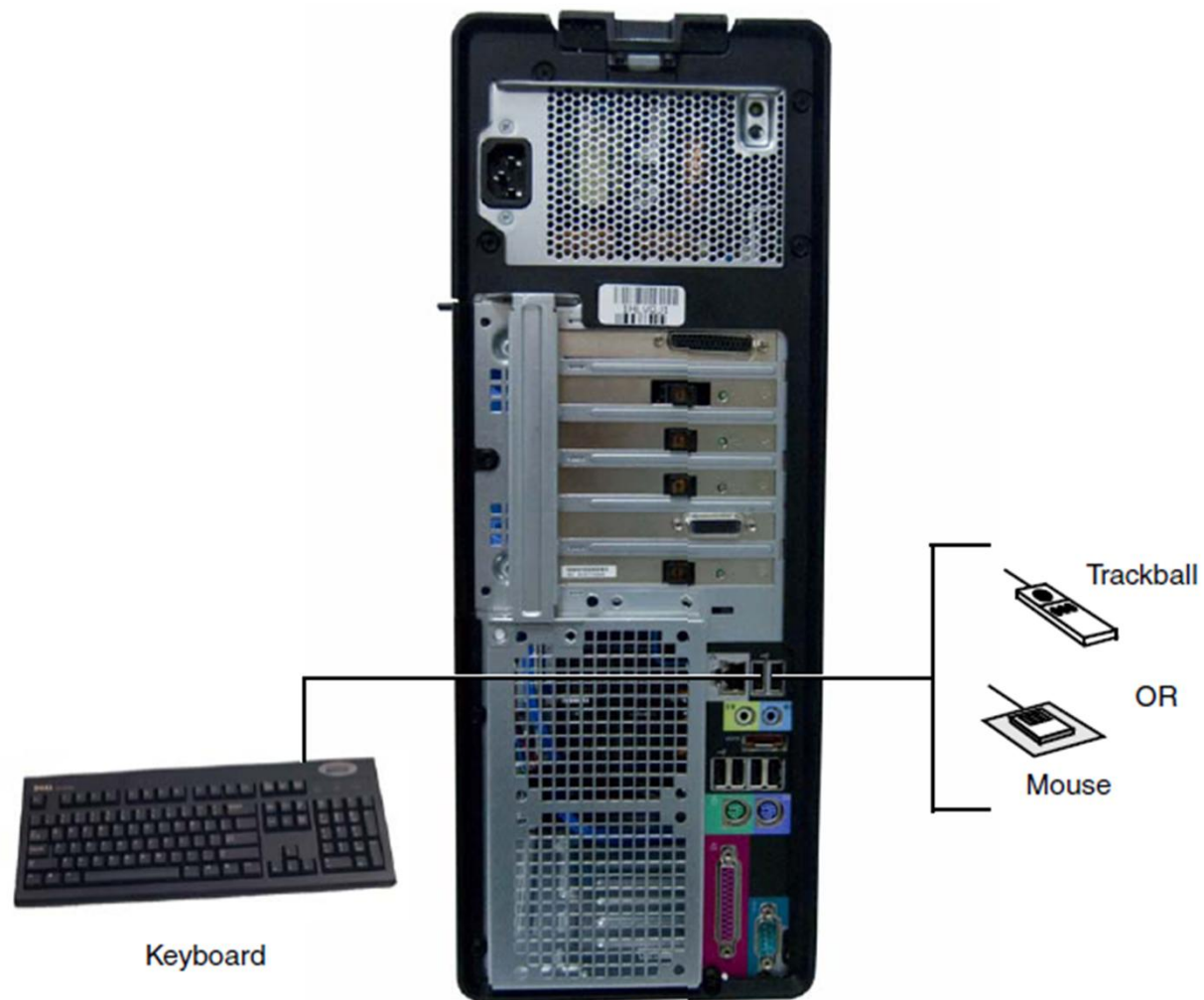
P92K Front Panel Layout (T3500)



Rear panel layout of P92K (T3500)



P92 Pointing Devices



P92 Printer Connections

- Alarm logging
- Reports
- Types
 - B/W dot-matrix 80
 - Color dot-matrix 132
 - Color PCL3
 - Color Postscript
- Cabling lengths
 - Dot-Matrix 80 & 132 – 7.5 m (25 ft)
 - Color printers – 4.5 m (15 ft) or 15 m (50 ft)

